

Quality Control in Chemical Industries
Quality can best be considered as the product of various attributes, including composition and Assthetic value, purity and consistency.
It should be thought of ^{as} their product rather than their sum, because if any one of these attributes is low the general quality will be low.
Broadly defined, quality helps to the ability of a product or service to consistently meet or exceed customer expectations.

Quality Control, in spite of variability in raw material due to its origin, means a system which standardises the materials used, the process itself and, in turn the product. Adjustments to the process should be made during manufacture. This requires the development of suitable rapid techniques of testing even though they may be indirect physical or empirical test methods.

⇒ In general therefore, Quality Control is a process that evaluates output relative to a standard and takes corrective action | measure when output does not meet standards.

⇒ The purpose of quality control is the manufacture of products that are standardised within approved - primarily selected commercial limits. Although, it is may ensure that overweight and very high quality goods are not sent out, it equally sees that the customer does not receive sub-standard products.

The organization of quality control is basically associated with the following sequence of operations:

* Raw material control

Examination of raw materials, most of which have already been partially processed and submitted to some form of standardisation.

- Formulation and examination of trial batches.
- Process control, which includes examination of intermediate diets and checking of recording charts on equipment and hygiene inspection.
- Product examination, including weight control and compliance with specification and legal requirements.

- ~~→ Storage control, including weight control and compliance with specification~~
- Storage control in the warehouse and at the retailer's premises.

Apart from these routine activities, the chemist also becomes involved in development work, the preparation of specifications and procedures, the evaluation of the comparative quality of competitors' products, the training of personnel and trouble-shooting.

The chief chemist usually acts as adviser on legislative matters and in view of the numerous statutory controls involved may sometimes have to 'turn down' some of the advertising department's attempts to use certain 'immediate' types of claims. In view of the importance of integration with other departments, it is most necessary for quality controllers to develop the ability to explain matters in terms that are understandable to associates who would otherwise be misled by scientific jargons.

The Main stages in Quality Control Systems

* Raw material control

The quality control of some products e.g food, begins in the farm. For instance, the best blends of wheat for bread-making and the most suitable fruits to use for fruit-juice making, canning and freezing are selected and inserted in specifications.

The firm can materially covers all items delivered to the manufacturer, whether used directly in the production or not. Apart from certain ingredients and water, it employs the quality control laboratory examines for instance, detergents, containers and even packaging materials, fuel and lubricating oils, and some times buildings materials and tools.

If the material is to come from sources of supply, a small buying sample is examined to ascertain that it has the desired properties for the intended product. Then the first delivery ordered and every subsequent one should correspond with that sample, unless the supplier gives a statement to the contrary.

The delivered materials are checked to see they are up to sample and in extreme cases the receiver can seek protection under the Sale of Goods Act.

More generally, a decision is made for every raw material received as to whether it is to be accepted, rejected, held pending correspondence and discussions, blended with other material or used for a different product. It should be borne in mind that such decisions (as with systems covering the sampling frequency) will depend to some extent on the good name (or otherwise) of the supplier in relation to past experience.

Raw materials are checked for identity and compliance with legal standards as to minimum quality, the absence of adulterants and limits of impurities, such as trace elements and other residues. Specifications, however, may require numerous other regular checks in order to standardise size, color, clarity, flavor, particle size and aroma, together with other factors that may affect the product quality and acceptability.

* Trial Batches

Trial batches of the product are prepared from certain raw materials, to check whether any recipe modification is required.

The correct metal to use for the equipment is also studied. Any product that contains acid, alcohol or salt is liable to corrode metals. The liability to corrosion can be checked before ordering the equipment by maintaining samples of the metal in the food at an elevated temperature.

The copper used for beer processing equipment has now largely been replaced by other metals such as aluminium, nickel and selected stainless steels.

* Ordinary stainless steel, which contains chromium and nickel together with small amounts of carbon, silicon and manganese, is not suitable for products that contain salt or vinegar. (Why is tin SD?)

* Process Control

Having checked that the raw materials are of good and suitable quality and commercially pure, it is necessary to consider whether it is advisable to carry out checks during the Process to enable certain modifications to be made.

Important considerations here are the best techniques to use, the processing times, temperatures and the relevant tools, instruments, gauges, meters and charts to check that the Procedure is standardised.

Rapid feed-back of information to the factory floor is all-important in Process Control, so lengthy methods of testing are inappropriate. During the Process, therefore, on-the-spot tests are devised when possible to measure composition and physical properties. For instance, Brix Refractometers measure the sugar content of syrups rapidly and enable quick adjustments to be made. Similarly, Salinometers or Brineometers give the salt concentration in brine.

There are also various detection systems in which magnets show up the presence of ferrous metals, rubber, stone and possibly glass. Such devices can be adjusted for varying sensitivity and the detector has to be linked with warning, rejection or Process-stopping system. Inspection devices are also available for detecting extraneous matter in glass containers.

It should be borne in mind that none of these devices represents a 100% safeguard and that they are fairly expensive to install.

Also, the factory lays down rules for Personnel, such as the following:

i, Containers closed with nails or wire should be opened in rooms away from the process area.

ib, Extraneous matter is removed by magnetic and electrostatic devices or by sieving.

iii, Machinery should be regularly examined for loose screws, nuts, bolts, washers and plating.

iv, No glass articles are allowed on the factory floor.

v, After maintenance, all loose materials must be taken away e.g wire, metal from drilling, solder and string.

vi, Overalls should have tie-on tapes instead of buttons, and no pockets.

vii, Women should be strongly discouraged from wearing clip-on jewellery i.e. no rings, bangles etc.

viii, Protective headgear is provided to prevent hair pins and hair from falling into products.

Such instructions have to be supplemented by some form of educational programme or publicity, as more important than rules is the condition of personnel into the idea of foreseeing possible troubles.

Product Examination

It is not truly a control. Nevertheless, Product examination should confirm that the desired results decided upon when the recipe was formulated have been achieved. For example, although ingredient purity has been ensured, there may have been contamination picked-up from plant during the process. Also, the ability to withstand storage can be confirmed only on the product itself. Therefore, representative samples are drawn from coded batches to confirm that the product complies with the specification and any legal or recognised limits. In addition, there is a considerable amount of psychological value in taking samples in full view of the factory personnel (even though samples may not be examined!).

A first consideration is weight. At the end of the production line, packets are weighed individually or go to check weights, which can be set to reject units at any selected level.

The weights should in all instances fall between the limits on the appropriate quality control charts.

Storage Control

This is mainly concerned with prediction of occurrences during the storage of the product. In other instances the life of the product can be assessed from accelerated storage tests in which the package, can, bottle or jar is submitted to the effect of elevated temperatures and sometimes humidity. The container and contents are then examined for faults, growth of organisms and insects, rancidity, loss of color etc.

The manufacturer must be concerned also with the way the product is stored after it has left the factory. This is done by producing leaflets of instructions to retailers, wholesalers and customers.

The Shopkeeper must be reminded of the importance of stock rotation on the "first-in, first-out" principle — Particularly Perishable ~~or~~ and frozen goods.

In essence, discrete checks should be made to ensure that the instructions are being passed 'down the line'.

Types of Samples.

— Complaint Samples:

These are samples usually brought back to the factory or reported to the factory by customers, or consumers alike due to some faults they found out on or about the product. These Complaints vary in type, but the most commonest types are those of off-taste, odour, extraneous matter etc.

The report on such samples should include any code or batch numbers, and if possible, positive advice should be given to try to prevent a recurrence of the problem/trouble.

— Competitor's Samples.

In a competitive market, information about products made by other manufacturers is always of interest to management. Competitor's samples are also valuable when one is preparing to start manufacturing a new product whose similarity is in the market — already. Organoleptic characteristics are important — as scientific tests and report should bring out comparisons with one's own product where applicable by tabulating the results of all samples on the same report.

Quality Assurance

In the context of quality assurance, there is a precise meaning of the word 'quality'. Here, it is defined as the aggregate of all features and characteristics of a Product/Service that bear on its ability to satisfy stated or implied needs. (ISO 8400-1986): Quality vocabulary.

Quality assurance encompasses all the technical and managerial aspects of product quality and safety during the design, specification, development, manufacturing and usage.

Periodic reviews of quality standards, and quality operations are also undertaken to ensure that products remain satisfactory to the consumer and competitive in the market place.

The points below describes exactly some of the attributes of quality assurance:

- 1, It is an aid to productivity.
- 2, It is good sense of management.
- 3, It is cost effective.
- 4, It is the responsibility of everyone.
- 5, It is the means of getting it right first time everytime.

Automation in quality Control

Automation is a system used in production or in quality control whereby a piece of equipment or device is given technical instructions or specifications to perform some tasks with little or no external assistance. Example, Viscometers, Pressure gauges, hydrometers etc. can be fitted into its production-line

to measure quantities according to specifications.

In some cases robots are used, these can reject or throw out intermediates or products outside its range or requirement which will then be taken out for correction or reification according to the desired specifications.

Automation is very useful and economical for continuous processes but difficult for batch operations/process due to stoppages or break in operations as a result of removal of products or other physical activity.

Regulatory Bodies

There are usually government established agencies empowered by law to regulate and enforce compliance to established standards in the production of goods and services, including import and export quantities of manufactured goods they also ensure that only quality products are introduced in a country's market or imported into the country. In Nigeria, we have two most prominent of these bodies SON (Standard Organisation of Nigeria) and NAFDAC (National Agency for Food, ~~Dr~~ and Drug Administration and Control).

These two bodies help ensure the quality and standard of goods both within and outside the country. While NAFDAC is primarily concerned with Food and drugs, SON on the other hand, ensures that standard manufacturing practices SMP or Good manufacturing Practices GMP of goods and services ranging from Engineering products, household, Industrial, clothing, Automobile, Electrical products, to mention but a few.

Types of Production Processes

(A). Batch Production Process: This is a process in which the reactants are mixed in a container or vessel and allowed to react and the product removed in a sequence of operations. Almost all laboratory-scale chemical processes are batch processes and are typically run at much lower temperatures and pressures. However, some commercial or industrial chemical processes still utilize batch processes or operations due to the nature of the product or demand for the products. Examples are: batch-wise full-boiled method for soap manufacture, batch-wise operations in drugs/pharmaceuticals/chemicals manufacture.

(B) Continuous Production Process: These are processes designed in such a way that as reactants are fed in, the products are removed. Many industrial-scale processes are of continuous process and are run at very high temperatures and pressures e.g. the Contact process, Haber process, Solvay process etc. Most continuous processes are fully or semi-automated operations.

Statistics in Quality Control

The application of statistics in quality control activities is known as statistical quality control (SQC). It is the use of statistical methods in monitoring and maintaining the quality of products and services. SQC is divided into two: Process control and product control.

⇒ Statistical process control method makes use of graphical displays called control charts to determine whether a process should be continued or adjusted to meet the set quality standards.